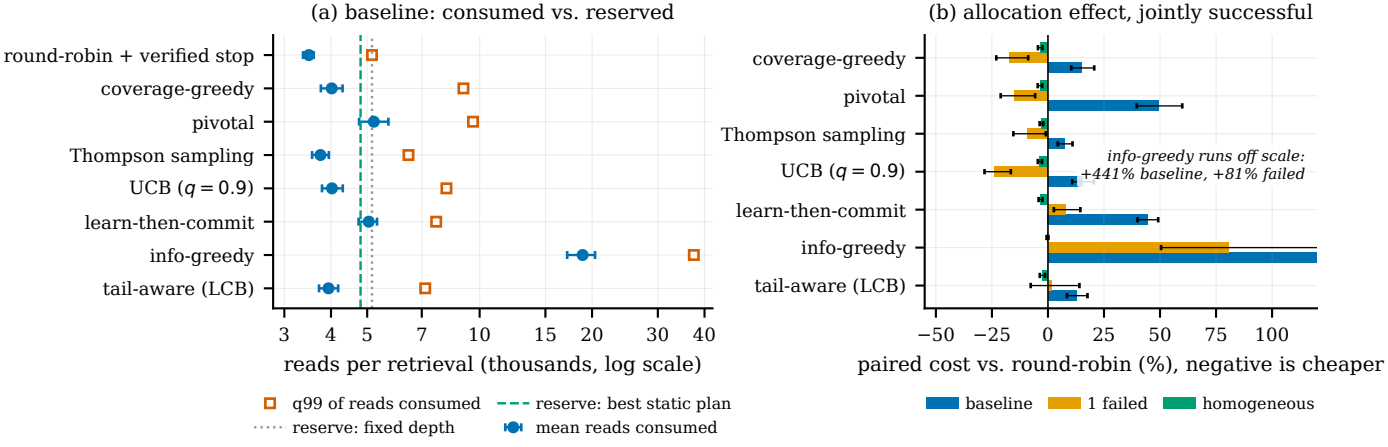
(a) baseline: consumed vs. reserved
round-robin + verified stop
coverage-greedy
pivotal
Thompson sampling
UCB ($q = 0.9$)
learn-then-commit
info-greedy
tail-aware (LCB)
3
4
5
7
10
15
20
30
40
reads per retrieval (thousands, log scale)
q99 of reads consumed
reserve: best static plan
reserve: fixed depth
mean reads consumed
(b) allocation effect, jointly successful
coverage-greedy
pivotal
Thompson sampling
UCB ($q = 0.9$)
learn-then-commit
info-greedy
tail-aware (LCB)
info-greedy runs off scale:
+441% baseline, +81% failed
−50
−25
0
25
50
75
100
paired cost vs. round-robin (%), negative is cheaper
baseline
1 failed
homogeneous